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Experiment No 9: Introduction to Network Simulation Tool – Basic Experiment

1. Aim

To study the basic features of a network simulation tool and create a simple network topology to observe packet transmission between network nodes.

2. Objective

- To understand the concept of network simulation.
- To become familiar with a GUI-based network simulator such as **Cisco Packet Tracer**.
- To create a simple network topology.
- To configure IP addresses on network devices.
- To test connectivity using **Ping**.
- To observe packet transmission through the network.

3. Software Required

- **Cisco Packet Tracer**
- Operating System: Windows/Linux

4. Theory

A **network simulator** is a software tool used to create and study computer networks without requiring physical networking hardware.

It allows us to create virtual:

- PCs
- Switches
- Routers
- Servers
- Network connections

We can configure these devices and observe how data packets travel through the network.

Cisco Packet Tracer is a GUI-based network simulation tool commonly used for learning and experimenting with computer networks.

5. Basic Network Topology

For this experiment, create the following topology:

```
   PC0
    |
    |
 Switch0
    |
    |
   PC1
```

Example IP addressing:

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

Both PCs belong to the same network:

Network: 192.168.1.0/24

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6. Procedure in Cisco Packet Tracer

Step 1: Open Packet Tracer

1. Open **Cisco Packet Tracer**.
2. Select **Network Devices** from the bottom-left panel.
3. Select **Switches**.
4. Drag a **2960 switch** into the workspace.

Step 2: Add PCs

1. Select **End Devices**.
2. Select **PC**.
3. Drag two PCs into the workspace.
4. Name them **PC0** and **PC1**.

Step 3: Connect the Devices

1. Click the **Connections** icon (lightning symbol).
2. Select **Copper Straight-Through** cable.
3. Connect:

```
PC0 FastEthernet0 → Switch0 FastEthernet0/1
PC1 FastEthernet0 → Switch0 FastEthernet0/2
```

Wait until the link indicators become green.

Step 7: Configure IP Address of PC0

1. Click **PC0**.
2. Select **Desktop**.
3. Select **IP Configuration**.
4. Enter:

```
IP Address:     192.168.1.1
Subnet Mask:    255.255.255.0
```

No default gateway is required for this simple same-network experiment.

Step 8: Configure IP Address of PC1

1. Click **PC1**.
2. Select **Desktop**.
3. Select **IP Configuration**.
4. Enter:

```
IP Address:     192.168.1.2
Subnet Mask:    255.255.255.0
```

---

9. Test Connectivity Using Ping

From PC0:

1. Click **PC0**.
2. Select **Desktop**.
3. Open **Command Prompt**.
4. Type:

```
ping 192.168.1.2
```

Press **Enter**.

You should get a response similar to:

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

This indicates that **PC0 can communicate successfully with PC1**.

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10. Observe Packet Transmission

Packet Tracer provides two modes:

Realtime Mode

The network operates normally and packets are transmitted immediately.

Simulation Mode

Simulation mode allows us to **observe packets step-by-step**.

To observe the packet:

1. Click **Simulation** at the bottom-right corner.
2. Select **Add Simple PDU** (envelope icon).
3. Click **PC0** as the source.
4. Click **PC1** as the destination.
5. Click **Capture/Forward**.

You can observe the packet travelling:

```
PC0 → Switch0 → PC1
```

The simulation can show protocols such as:

- ARP
- ICMP

For a ping operation, **ICMP** is used to test connectivity. Before the first ping, **ARP** may be used to find the MAC address corresponding to PC1's IP address.

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11. Observation

Parameter	Observation
Number of PCs	2
Number of switches	1
Network	192.168.1.0/24
PC0 IP	192.168.1.1
PC1 IP	192.168.1.2
Connection	Successful
Protocol observed	ARP and ICMP
Packet path	PC0 → Switch → PC1

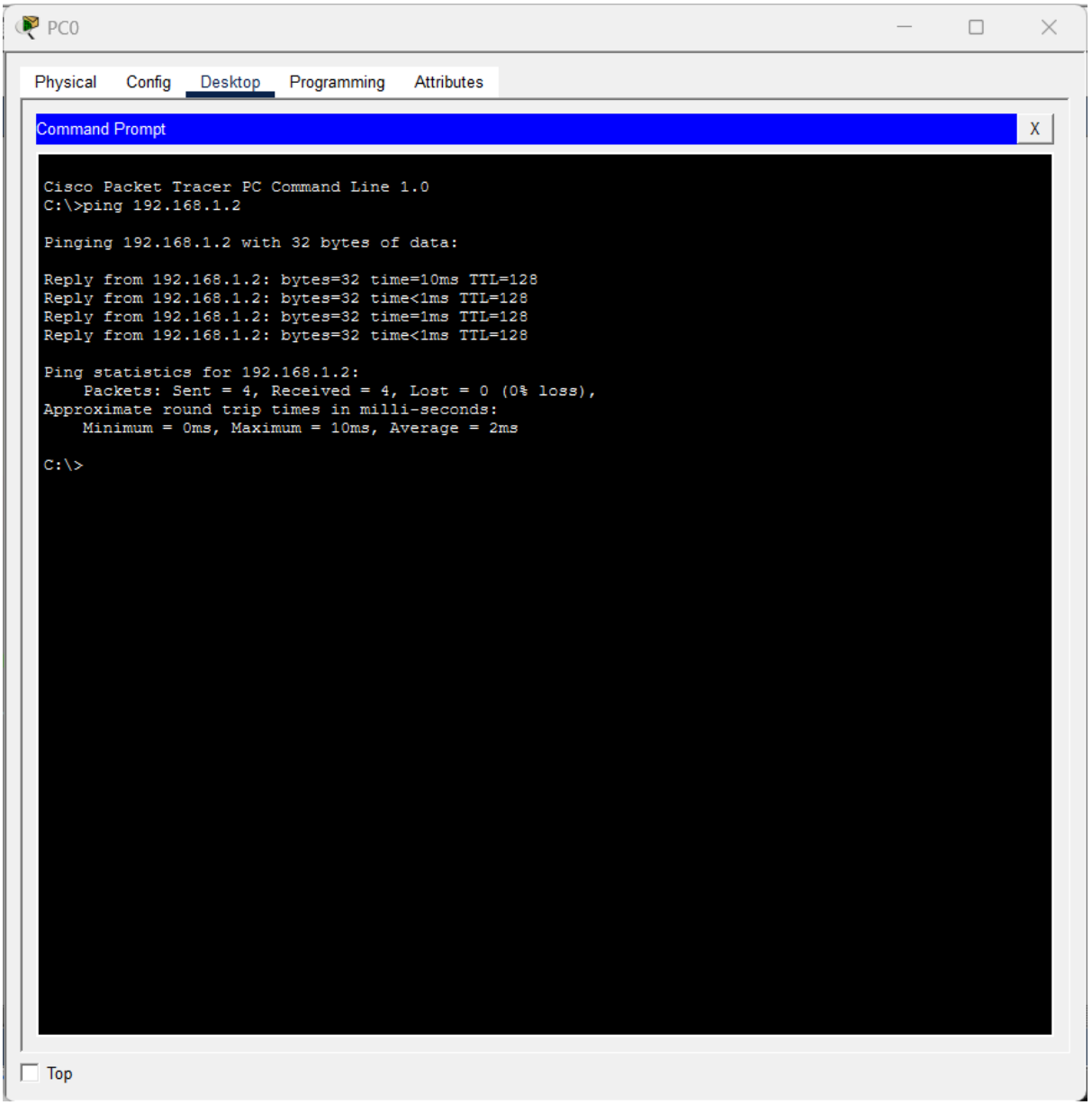
12. Result

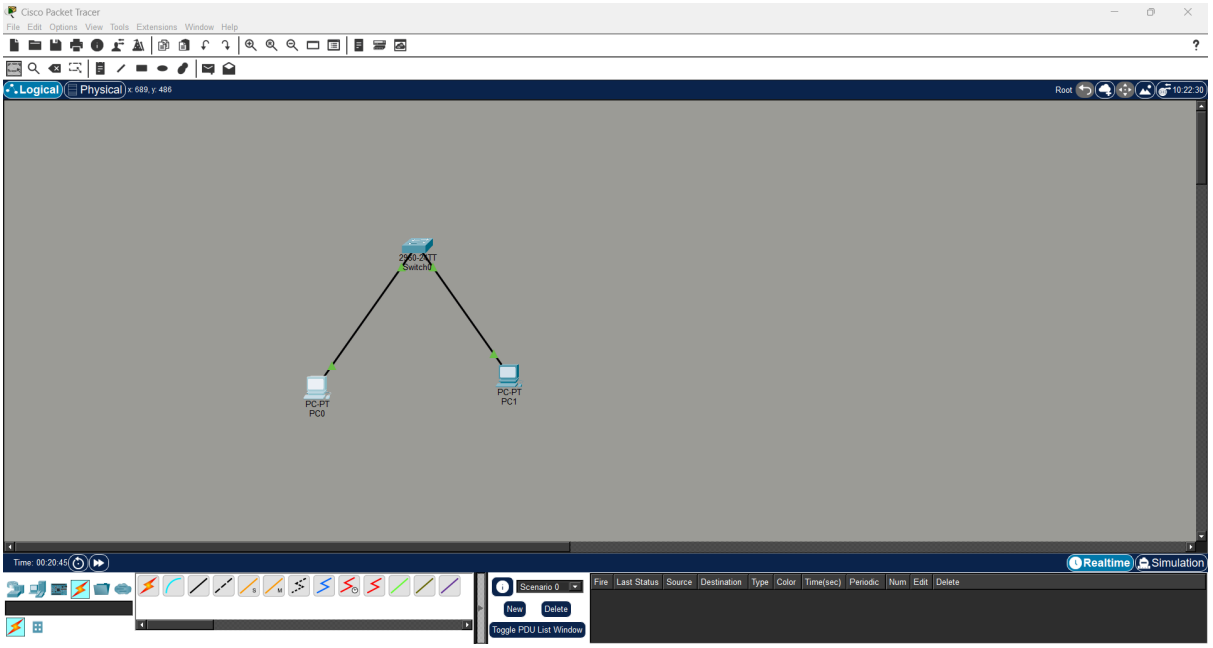
The basic features of **Cisco Packet Tracer**, a GUI-based network simulation tool, were studied successfully. A simple network consisting of two PCs and a switch was created, IP addresses were configured, and connectivity was successfully verified using the **Ping command**. Packet transmission was also observed using **Simulation Mode**.

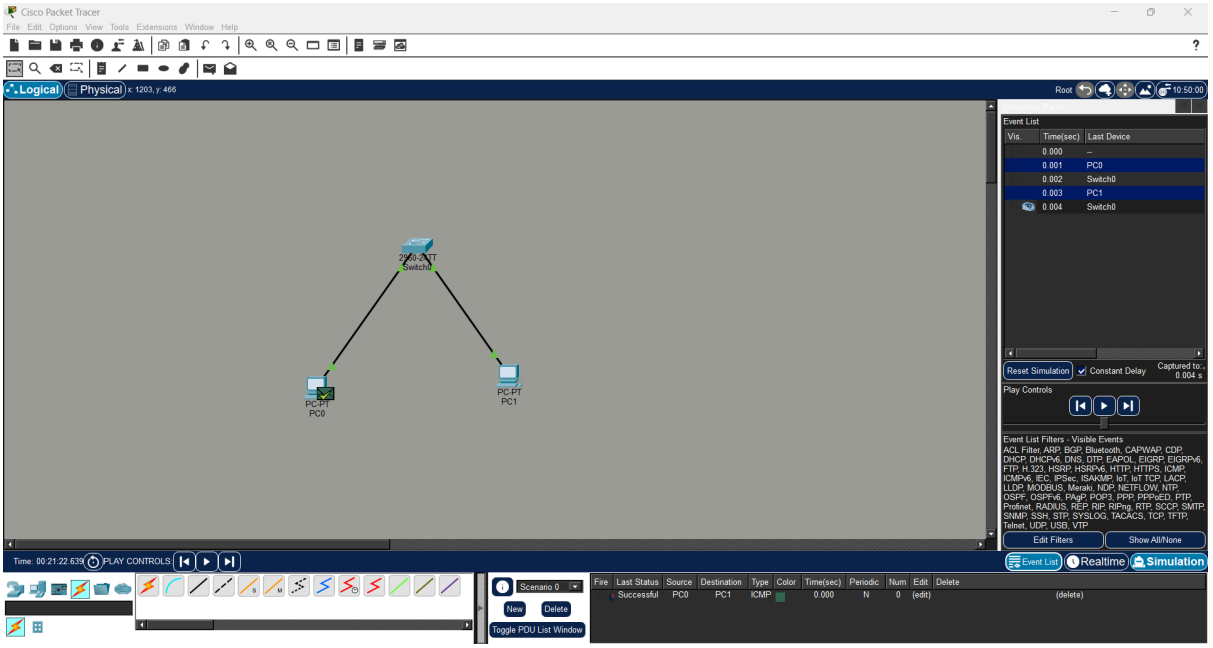
13. Conclusion

Cisco Packet Tracer provides a convenient environment for designing, configuring, and analyzing computer networks without physical hardware. It helps students understand network communication, IP addressing, ARP, ICMP, switching, and packet flow through practical simulation.

14. Screenshots




Cisco Packet Tracer
Logical
Physical
Switch0
PC-PT PC0
PC-PT PC1
Realtime
Simulation


Cisco Packet Tracer
Logical
Physical
Switch0
PC-PT PC0
PC-PT PC1
Event List
Reset Simulation
Constant Delay
Play Controls
Realtime
Simulation